

Validators Class Documentation

This document explains the `Validators` class used for form validation in Flutter. You can review, edit, and export this document as needed.

Overview

The `Validators` class provides reusable static methods to validate common form fields such as name, email, password, confirm password, and phone number. Each method returns a `String` error message if validation fails, or `null` if the input is valid.

This pattern is commonly used with Flutter `TextFormField` widgets.

Class Definition

```
class Validators {
  static String? name(String? value) {
    if (value == null || value.trim().isEmpty) {
      return 'Name is required';
    }
    if (value.trim().length < 3) {
      return 'Name must be at least 3 characters';
    }
    return null;
  }

  static String? email(String? value) {
    if (value == null || value.trim().isEmpty) {
      return 'Email is required';
    }
    final emailRegex =
      RegExp(r'^[\w-\.\.]+\@([\w-]+\.\.)+[\w-]{2,4}$');
    if (!emailRegex.hasMatch(value)) {
      return 'Enter a valid email';
    }
    return null;
  }

  static String? password(String? value) {
    if (value == null || value.isEmpty) {
      return 'Password is required';
    }
  }
}
```

```

        if (value.length < 6) {
            return 'Password must be at least 6 characters';
        }
        return null;
    }

    static String? confirmPassword(
        String? value,
        String password,
    ) {
        if (value == null || value.isEmpty) {
            return 'Confirm password is required';
        }
        if (value != password) {
            return 'Passwords do not match';
        }
        return null;
    }

    static String? phone(String? value) {
        if (value == null || value.trim().isEmpty) {
            return 'Phone number is required';
        }
        if (!RegExp(r'^\d{10}$').hasMatch(value)) {
            return 'Enter a valid 10-digit number';
        }
        return null;
    }
}

```

Method Details

1. Name Validator

Method: `Validators.name`

- Ensures the name is not empty
- Requires at least 3 characters

Error Messages: - `Name is required` - `Name must be at least 3 characters`

2. Email Validator

Method: `Validators.email`

- Checks if email is provided
- Validates format using regular expression

Error Messages: - `Email is required` - `Enter a valid email`

3. Password Validator

Method: `Validators.password`

- Ensures password is not empty
- Minimum length of 6 characters

Error Messages: - `Password is required` - `Password must be at least 6 characters`

4. Confirm Password Validator

Method: `Validators.confirmPassword`

- Ensures confirmation is not empty
- Matches with original password

Parameters: - `value`: Confirm password input - `password`: Original password

Error Messages: - `Confirm password is required` - `Passwords do not match`

5. Phone Number Validator

Method: `Validators.phone`

- Ensures phone number is not empty
- Accepts exactly 10 digits

Error Messages: - `Phone number is required` - `Enter a valid 10-digit number`

Usage Example

```
TextFormField(  
  validator: Validators.email,  
)
```

Notes

- All methods are `static`, so no instance is required
- Designed for use with Flutter forms
- Easily extendable for additional validation rules

You can now edit this document or export it as PDF/Word if needed.